

Mathematics

Stage 9

Paper 1

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Geometrical instruments
Tracing paper (optional)

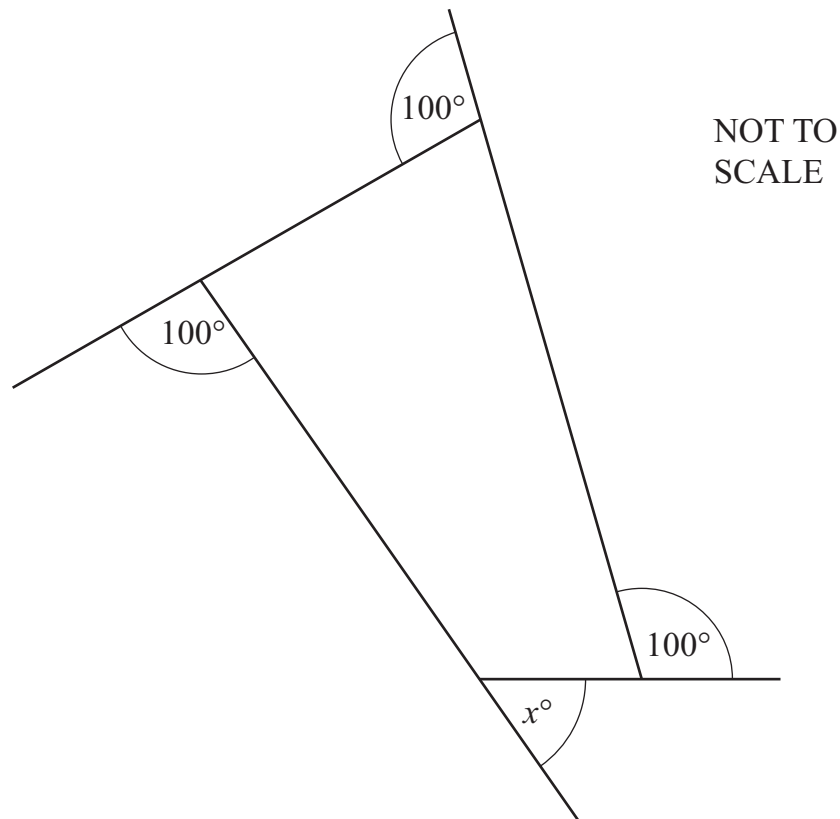
INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You are **not** allowed to use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 The diagram shows a quadrilateral with the exterior angles marked.



Calculate the value of x .

$x = \dots\dots\dots$ [1]

- 2 Here are some calculations.

-9×7

$-8 \div (-4)$

$(-3)^2$

$12 \div (-3)$

Write each calculation in the correct column in the table.

One has been done for you.

Answer is positive	Answer is negative
	-9×7

[1]

- 3 The term-to-term rule of a sequence is **square and then add 2**
The 1st term of the sequence is 3

Find the 2nd term of the sequence.

..... [1]

- 4 Draw a ring around the number that is rational.

$\sqrt{2}$

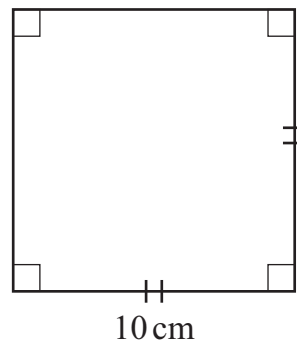
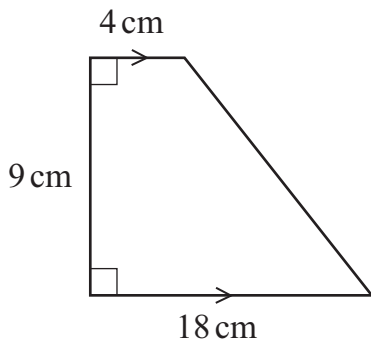
$\sqrt{4}$

$\sqrt{6}$

$\sqrt{8}$

[1]

- 5 Show that the area of the trapezium is smaller than the area of the square.



NOT TO
SCALE

[2]

- 6 The value of x is an integer.

$$9 < x + 4 < 12$$

Write down the two possible values of x .

..... or [1]

- 7 Here are the equations of some straight-line graphs.

$$y = 2x + 3$$

$$y = -2x$$

$$y = 2x$$

$$y = -2x + 5$$

Write each equation in the correct place in the table.

One has been done for you.

	Gradient is positive	Gradient is negative
Passes through (0, 0)		
Does not pass through (0, 0)	$y = 2x + 3$	

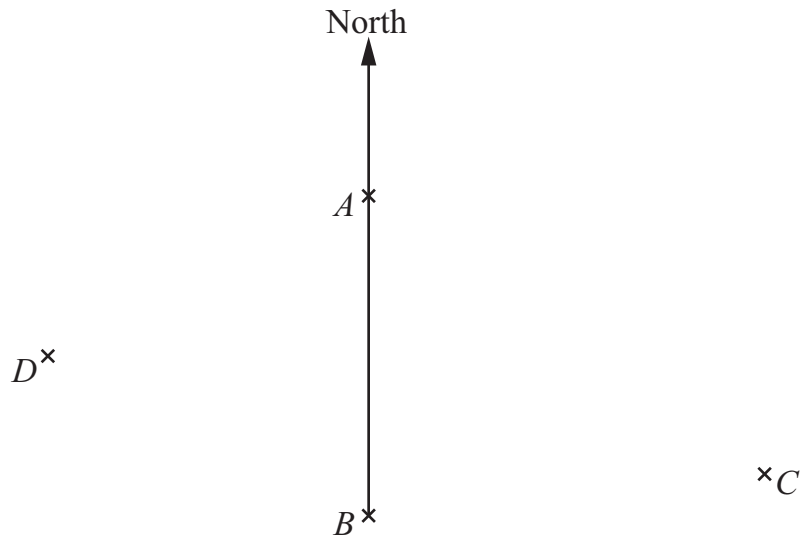
[1]

- 8 Find the 5th term for each of the sequences in the table.

n th term rule	5th term
$\frac{n}{4}$	
$n^2 - 11$	
n^3	

[2]

- 9 The diagram shows the positions of points A , B , C and D .



Draw a line to join each bearing to the correct description.

Bearing of B from A	Less than 180°
Bearing of C from B	180°
Bearing of D from B	More than 180°

[1]

- 10 Write a number in each box to complete the calculation.

$$3.5 \div \frac{7}{8} = \frac{\boxed{}}{2} \times \frac{\boxed{}}{7} = \boxed{}$$

[1]

11 The data shows the marks scored by students in a test.

45	46	47	49	50	50	52	53
54	55	56	57	57	58	59	59
61	62	64	65	66	72	73	74

Jamila records the test marks in this table.

Test mark	Frequency
20–39	0
40–59	16
60–79	8
80–99	0

(a) Explain why the class intervals chosen by Jamila are **not** the most appropriate.

.....
 [1]

(b) Complete the frequency table to record the data using more appropriate class intervals.
 You should choose class intervals with equal widths.
 You may not need all the rows in the table.

Test mark	Frequency

[2]

12 Simplify.

$$(a^3)^4$$

.....

$$\frac{4n+6}{2}$$

.....

[2]

13 (a) Write 50 000 in standard form.

..... [1]

(b) Write 4.07×10^{-3} as an ordinary number.

..... [1]

14 The counters in a bag are either red or blue or green or yellow.

A counter is picked at random from the bag.

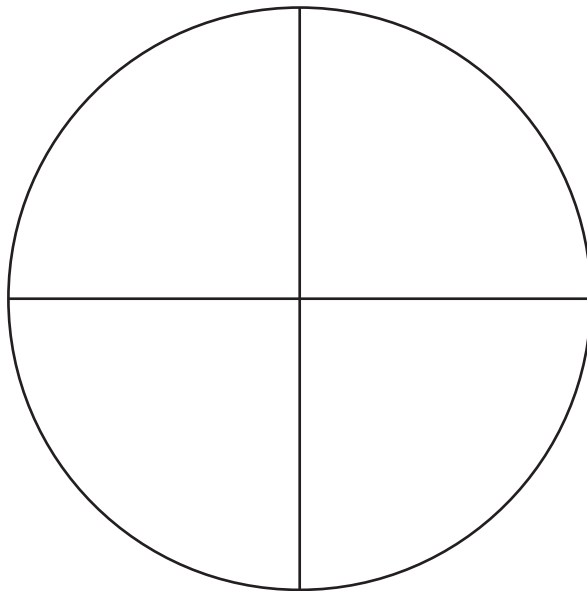
The table shows the probability of some of the outcomes.

Colour of counter	Red	Blue	Green	Yellow
Probability	0.15	0.05	0.35	

Find the probability that the counter is red or yellow.

..... [2]

- 15** The diagram shows a circle with a horizontal diameter and a vertical diameter drawn.



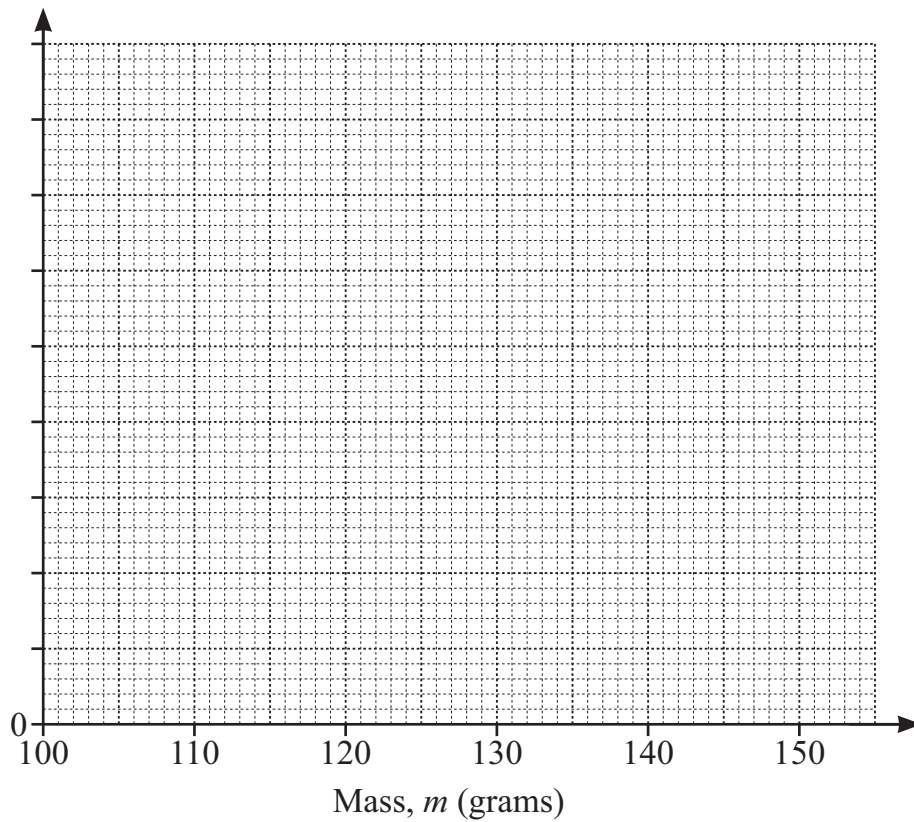
Use the diagram to construct an inscribed regular octagon.
Do not rub out your construction arcs.

[2]

16 Naomi measures the mass of each of 50 bananas.

Mass, m (grams)	Frequency
$100 \leq m < 110$	12
$110 \leq m < 120$	15
$120 \leq m < 130$	17
$130 \leq m < 140$	4
$140 \leq m < 150$	2

(a) Draw a frequency polygon to show this information.



[3]

(b) Draw a ring around the class interval that contains the median mass.

$$100 \leq m < 110$$

$$110 \leq m < 120$$

$$120 \leq m < 130$$

$$130 \leq m < 140$$

[1]

17 Find the value of

$$10^3 \times 29 \times 10^{-3}$$

..... [1]

18 The values of a , b and c are $a = 10$ $b = 7$ $c = 4$

Tick (✓) to show if the value of each of these expressions is equal to 64 or **not** equal to 64

	Equal to 64	Not equal to 64
$6(a + c)$	☐	☐
$\frac{a^2 + 28}{2}$	☐	☐
$(6 - 2b)^2$	☐	☐
$b^2 + c^2 - 1$	☐	☐

[2]

19 Calculate.

$$3\frac{1}{4} - \left(1\frac{1}{2} + 1\frac{1}{3}\right)$$

..... [3]

20 Write the correct power in each box.

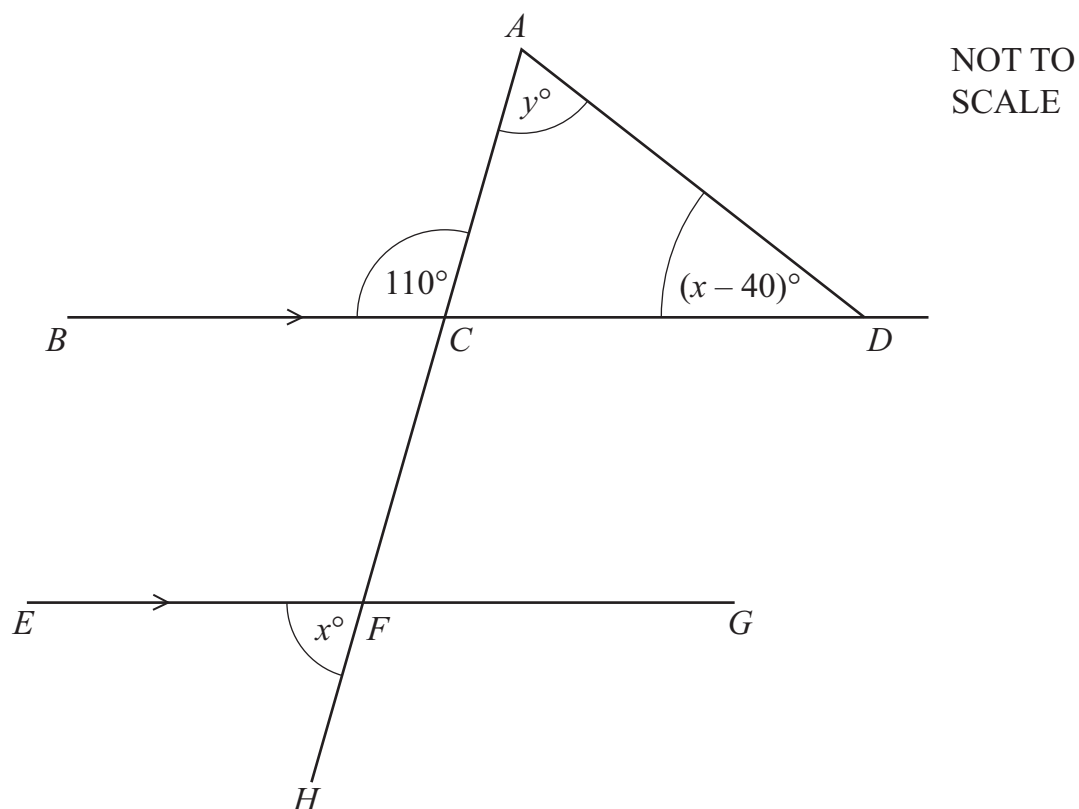
$$6^5 \times 6^{-1} = 6 \boxed{}$$

$$8^4 \div 8^{-2} = 8 \boxed{}$$

$$\frac{1}{81} = 9 \boxed{}$$

[2]

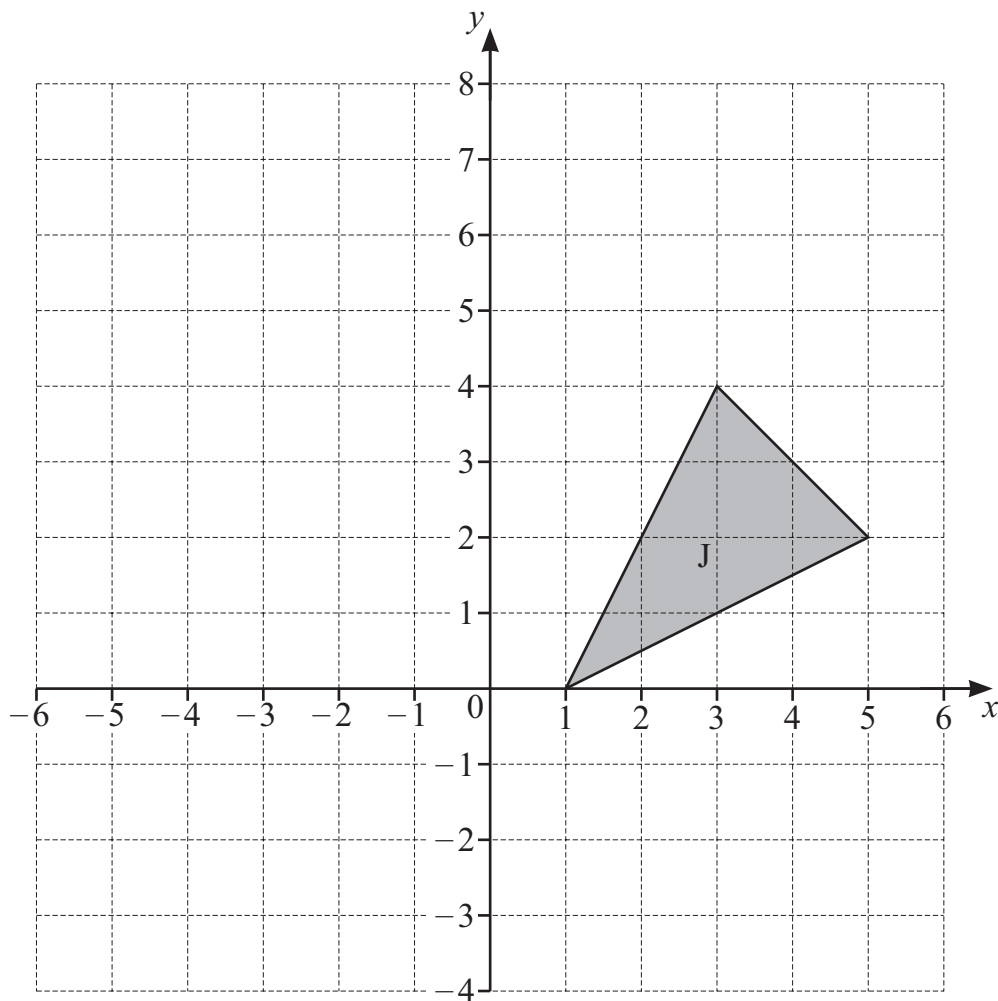
- 21 BCD and EFG are parallel lines.
 $ACFH$ is a straight line.



Calculate the value of y .

$y =$ [2]

22 Triangle J is shown on the grid.



Triangle J is reflected in the line $y = 3$ to give triangle K.

Triangle K is then rotated by 90° anticlockwise, centre $(0, 2)$, to give triangle L.

Draw and label triangle L on the grid.

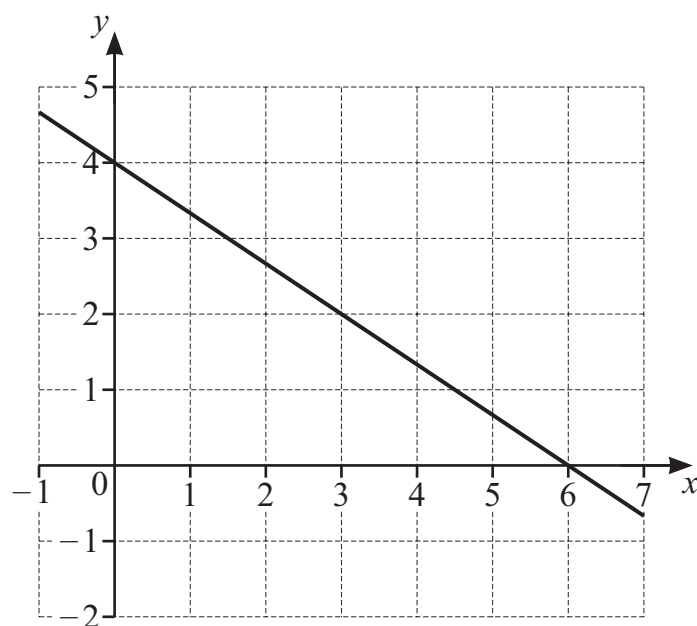
[2]

23 Tick (✓) to show if each statement is true or false.

	True	False
$2 + 3 \times \sqrt{36} = 30$	☐	☐
$\sqrt[3]{8 \times 3 + 3} = 9$	☐	☐

[1]

24 The diagram shows the graph of the line $2x + 3y = k$.



Draw a ring around the value of k .

4

6

12

24

[1]

25 Calculate.

$$\frac{0.8^2 - 0.01}{-0.09}$$

..... [3]

26 The values of x and y satisfy these simultaneous equations.

$$y = 4x - 2$$

$$3y = 7x + 14$$

Find the value of $2x - y$.

..... [4]

- 27 Chen and Hassan each play a game.
They can each either win or lose or draw the game.

Chen	Hassan
Probability Chen wins is 0.3 Probability Chen draws is 0.1	Probability Hassan wins is 0.5 Probability Hassan draws is 0.2

The outcome of Chen's game is independent of the outcome of Hassan's game.

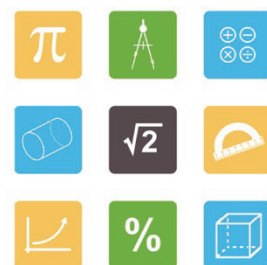
Find the probability that Chen and Hassan **both** lose their games.

..... [2]

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Mathematics

Stage 9

Paper 1

2024

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**General guidance on marking****Mark scheme annotations and abbreviations**

FT	follow through after error
dep	dependent
oe	or equivalent
cao	correct answer only
isw	ignore subsequent working
soi	seen or implied

Difference in printing

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Square brackets in mark scheme

When square brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6[58...]	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30..... cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e.18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

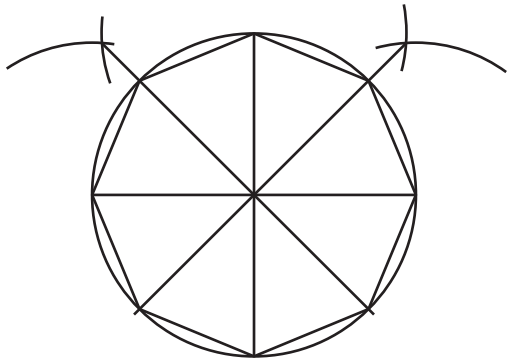
The table shows acceptable and unacceptable ways to plot points.

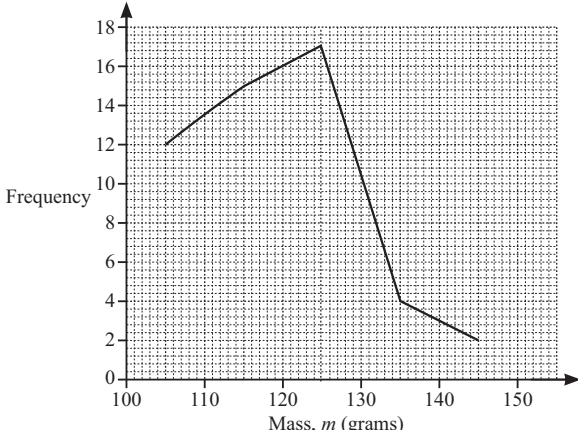
Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance									
1	(x =) 60	1											
2	<table><tr><td>Answer is positive</td><td>Answer is negative</td></tr><tr><td>$-8 \div (-4)$</td><td>-9×7</td></tr><tr><td>$(-3)^2$</td><td>$12 \div (-3)$</td></tr></table>	Answer is positive	Answer is negative	$-8 \div (-4)$	-9×7	$(-3)^2$	$12 \div (-3)$	1		All three answers correct for the mark. Answers can be in either order within the correct column.			
Answer is positive	Answer is negative												
$-8 \div (-4)$	-9×7												
$(-3)^2$	$12 \div (-3)$												
3	11	1											
4	$\sqrt{4}$	1		Accept any clear indication.									
5	(Area of trapezium =) $99 \text{ (cm}^2\text{)}$ and (Area of square =) $100 \text{ (cm}^2\text{)}$ seen	2	Award 1 mark for correct method for finding the area of the trapezium, e.g. $0.5 \times (4 + 18) \times 9$ $4 \times 9 + 0.5 \times (18 - 4) \times 9$	1 mark can be implied by sight of 99									
6	6 and 7	1		Both answers correct in either order for the mark. Do not accept any additional values.									
7	<table><tr><td></td><td>Gradient is positive</td><td>Gradient is negative</td></tr><tr><td>Passes through (0, 0)</td><td>$y = 2x$</td><td>$y = -2x$</td></tr><tr><td>Does not pass through (0, 0)</td><td>$y = 2x + 3$</td><td>$y = -2x + 5$</td></tr></table>		Gradient is positive	Gradient is negative	Passes through (0, 0)	$y = 2x$	$y = -2x$	Does not pass through (0, 0)	$y = 2x + 3$	$y = -2x + 5$	1		All three answers correct for the mark.
	Gradient is positive	Gradient is negative											
Passes through (0, 0)	$y = 2x$	$y = -2x$											
Does not pass through (0, 0)	$y = 2x + 3$	$y = -2x + 5$											
8	$\frac{5}{4}$ or 1.25 14 125	2	Award 1 mark for two correct values.	Accept equivalent fractions for $\frac{5}{4}$ Do not accept answers $5^2 - 11$ or 5^3									

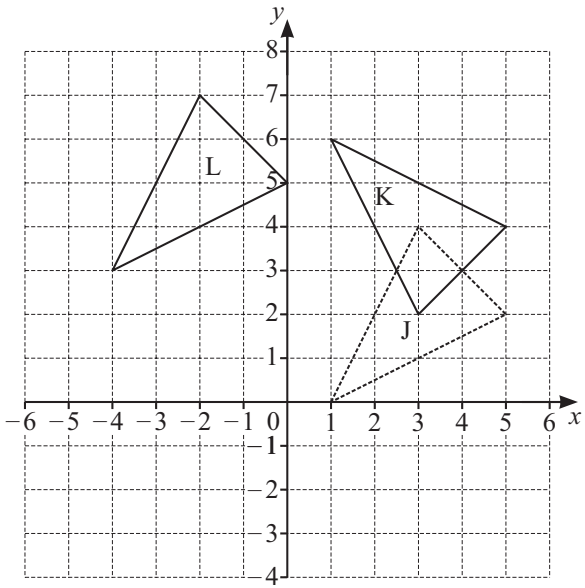
Question	Answer	Marks	Part Marks	Guidance
9	Bearing of B from A Less than 180° Bearing of C from B 180° Bearing of D from B More than 180°	1		All three lines correct for the mark.
10	$3.5 \div \frac{7}{8} = \frac{\boxed{7}}{2} \times \frac{\boxed{8}}{7} = \boxed{4}$	1		Accept equivalent correct answers in the first two boxes.

Question	Answer	Marks	Part Marks	Guidance																				
11(a)	A correct explanation, e.g. - Only 2 class intervals have non-zero frequencies - All the data are between 45 to 74 (but the intervals cover 20 to 99).	1		Accept - She should have more class intervals - The first and last class intervals are not used.																				
11(b)	A correctly completed frequency table based on a set of class intervals that - cover the data - have equal width <16 - do not overlap - have at least 3 non-zero frequencies e.g. <table><tr><td>45–49</td><td>4</td></tr><tr><td>50–54</td><td>5</td></tr><tr><td>55–59</td><td>7</td></tr><tr><td>60–64</td><td>3</td></tr><tr><td>65–69</td><td>2</td></tr><tr><td>70–74</td><td>3</td></tr></table> or <table><tr><td>40–49</td><td>4</td></tr><tr><td>50–59</td><td>12</td></tr><tr><td>60–69</td><td>5</td></tr><tr><td>70–79</td><td>3</td></tr></table>	45–49	4	50–54	5	55–59	7	60–64	3	65–69	2	70–74	3	40–49	4	50–59	12	60–69	5	70–79	3	2	Award 1 mark for a suitable set of class intervals or for finding the correct frequencies for <i>their</i> frequency table (<i>their</i> frequencies should sum to 24 and at least 3 should be non-zero).	
45–49	4																							
50–54	5																							
55–59	7																							
60–64	3																							
65–69	2																							
70–74	3																							
40–49	4																							
50–59	12																							
60–69	5																							
70–79	3																							
12	a^{12} $2n + 3$	2	Award 1 mark for each correct answer.																					

Question	Answer	Marks	Part Marks	Guidance
13(a)	5×10^4	1		
13(b)	0.00407	1		
14	0.6 or equivalent	2	Award 1 mark for $P(\text{yellow}) = 0.45$ or equivalent or $1 - 0.05 - 0.35$ or equivalent or <i>their</i> $P(\text{yellow}) + 0.15$ provided the answer is < 1	0.45 may be seen in table.
15	A correct construction of a regular octagon, with correct construction arcs, e.g. 	2	Award 1 mark for constructing the bisector of one right angle.	Accept equivalents, e.g. constructing perpendicular bisectors of lines joining ends of diameters.

Question	Answer	Marks	Part Marks	Guidance
16(a)	A correct graph with axes clearly numbered and labelled, e.g. 	3	Award 2 marks for a correct frequency polygon but without the vertical axis numbered or labelled or for at least four correctly plotted points and a numbered and labelled vertical axis. or Award 1 mark for plotting at least four points with the correct vertical plots or the correct horizontal plots or for numbering and labelling the vertical axis suitably.	Ignore lines to the left of 105 or to the right of 145 For 3 marks, the vertical axis should be numbered and labelled suitably. - Vertical axis should be numbered to at least 16 - Accept different scales (e.g. 1 big square = 5) - Label could be '(Number of) bananas' For 1 mark, the vertical axis may not be numbered if the horizontal plots are correct.
16(b)	$110 \leq m < 120$	1		Accept any clear indication.
17	29	1		
18	☐ ☒ ☒ ☐ ☒ ☐	2	Award 1 mark for three correct ticks.	Accept any clear indication.

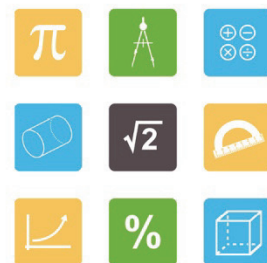
Question	Answer	Marks	Part Marks	Guidance
19	$\frac{5}{12}$	3	Award 2 marks for sight of $2\frac{5}{6}$ or $1\frac{3}{4}$ or $1\frac{11}{12}$ or for $1\frac{1}{4} - \frac{5}{6}$ or equivalent or $\frac{39}{12} - \frac{18}{12} - \frac{16}{12}$ or equivalent or better or $\frac{3}{12} - \frac{6}{12} - \frac{4}{12} + 1$ or equivalent or better. or Award 1 mark for converting two fractions correctly to a common denominator or converting any mixed number correctly to an improper fraction.	Accept equivalent fractions for all marks.
20	4 6 -2	2	Award 1 mark for one or two correct powers.	

Question	Answer	Marks	Part Marks	Guidance
21	$(y =) 80$	2	Award 1 mark for $(x =) 70$ or if <i>their</i> $y + \text{their } x = 150$ or $y + x - 40 = 110$	1 mark can be implied by angle $ADC = 30(^{\circ})$ (check diagram)
22	Showing L correctly drawn at $(-4, 3)$, $(0, 5)$, $(-2, 7)$ 	2	Award 1 mark for finding the position of K correctly or for applying the correct rotation to <i>their</i> K, provided it fits on the grid.	Accept labels not shown, provided the intention is clear.

Question	Answer	Marks	Part Marks	Guidance
23	☐ ☒ ☐ ☒	1		Both answers correct for the mark. Accept any clear indication.
24	12	1		Accept any clear indication.
25	-7	3	Award 2 marks for $\frac{63}{[-]9}$ or for an answer containing the digit $[-]7$ or Award 1 mark for 0.63 seen or for $\frac{\text{their } 0.63 \times 100}{[-]9}$ or better.	e.g. -0.07, 70 or 7

Question	Answer	Marks	Part Marks	Guidance
26	-6	4	Award 3 marks for $x = 4$ and $y = 14$ or for $x = 4$ or $y = 14$ and correctly finding <i>their</i> $(2x - y)$ or $x = 4$ and $2x - y = 2x - (4x - 2)$ or better. or Award 2 marks for $x = 4$ or $y = 14$ or Award 1 mark for a correct method to solve the simultaneous equations or for getting a value of x and y that satisfy one of the equations or for finding <i>their</i> $(2x - y)$.	A correct method could involve substituting the first equation into the second equation or multiplying one or both equations by constants followed by a correct decision about whether to add or subtract.
27	0.18	2	Award 1 mark for 0.6 and 0.3 seen or $(1 - (0.3 + 0.1)) \times (1 - (0.5 + 0.2))$	0.6 and 0.3 may be seen on a tree diagram.

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Mathematics

Stage 9

Paper 2

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour

Additional materials: Calculator
Geometrical instruments
Tracing paper (optional)

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should show all your working on the question paper.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

- 1 Draw a ring around the sum of the interior angles in a hexagon.

180°

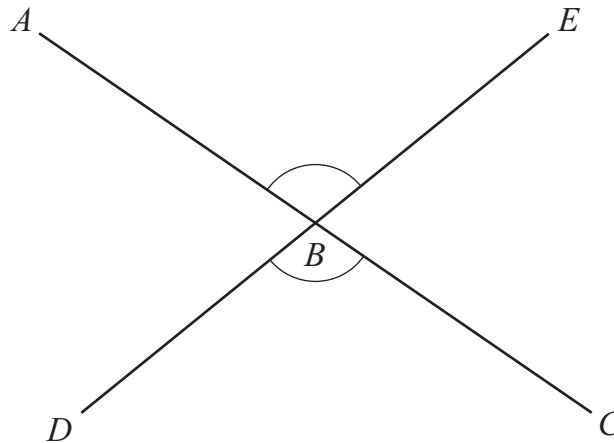
360°

720°

1080°

[1]

- 2 The diagram shows two straight lines, ABC and DBE .
 ABC and DBE are **not** perpendicular.

NOT TO
SCALE

Tick (✓) to show if each of these statements is true or false.

	True	False
ABE and DBC are corresponding angles.	☐	☐
Angle $ABE = \text{angle } DBC$.	☐	☐
Angle $ABE + \text{angle } DBC = 180^\circ$.	☐	☐

[1]

- 3 The back-to-back stem-and-leaf diagram shows the ages of some of the people in two choirs.

Choir A					Choir B				
				2	3	5			
		5	3	3	1	3	4	4	9
7	4	1	0	4	2	5	6	7	7
	9	6	2	5	3	4	8		
			3	6	0	7			
		5	1	7					

Key: 2 | **5** | 3 represents a person aged 52 years in Choir A and a person aged 53 years in Choir B

The ages of four people have not been included in the diagram.

Choir A	Choir B
Pierre is 28 years old Mike is 29 years old	Anastasia is 68 years old Samira is 72 years old

Complete the diagram by entering the ages of these four people.

[2]

- 4 Here are two properties about a number x .

x is greater than 344

x rounds to 340 correct to 2 significant figures.

Write down a possible value of x .

..... [1]

5 Solve.

$$\frac{18}{y} = 3$$

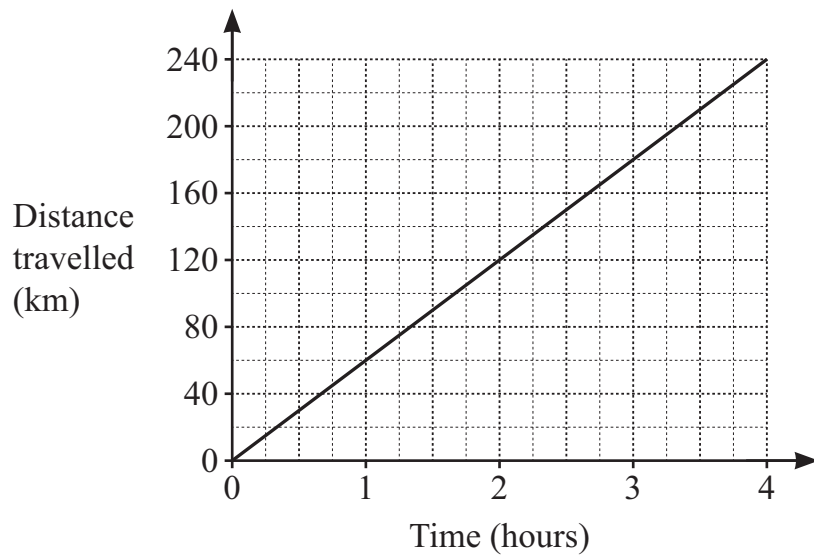
$$y = \dots\dots\dots [1]$$

6 A is the point $(1, 4)$ and B is the point $(1, 10)$.

Find the coordinates of the point one third of the way along AB from A .

$$(\dots\dots\dots, \dots\dots\dots) [1]$$

7 Angelique's journey is represented in the distance–time graph.



Calculate Angelique's speed on her journey.

$$\dots\dots\dots \text{ km/h } [1]$$

8 Here are some symbols.

< > =

Complete each statement by writing one of the symbols.

1 light year 1000 km

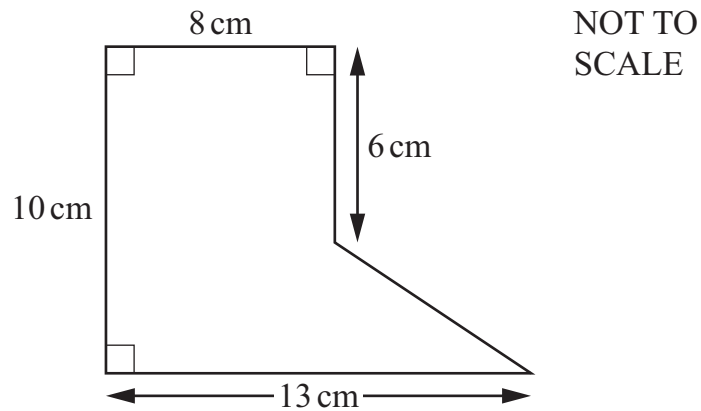
1 tonne 1000 kg

1 microgram 1000 grams

1 terabyte 1000 bytes

[2]

9 The diagram shows the cross-section of a prism.



The prism has a length of 7 cm.

Calculate the volume of the prism.

..... cm³ [2]

- 10** An old coin has a value of \$5000
The value of the coin increases by 2% every year.

Calculate the value of the coin after 3 years.

\$ [2]

- 11** Write a number in each box to complete each expansion.

$$(x - 3)(x + 11) = x^2 + \boxed{}x - 33$$

$$(x + 7)^2 = x^2 + \boxed{}x + 49$$

[2]

- 12** Ahmed wants to find out how much time students in his school spend on their homework.
He decides to choose 12 students from his class as his sample.
Safia says, ‘Ahmed could improve his sample by making some changes.’

Tick (✓) each of the changes that should give Ahmed a better sample.

Choose more students.

☐

Choose students from different classes.

☐

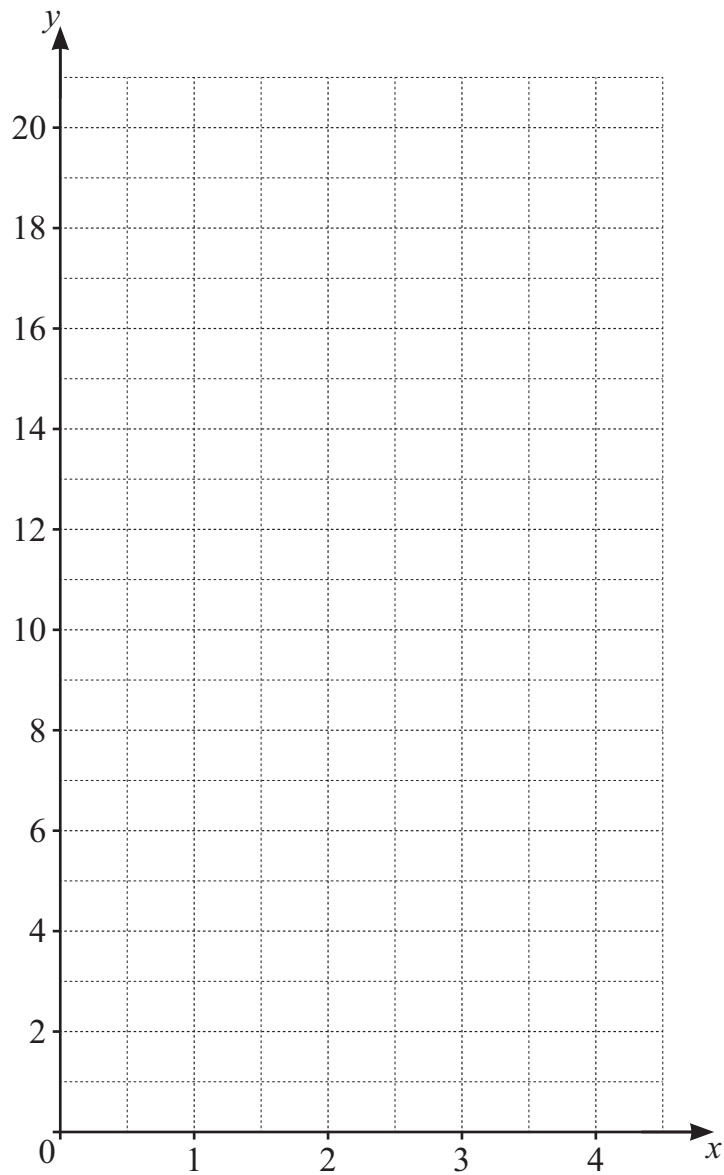
Choose students from different schools.

☐

[1]

- 13** Draw the graph of $y = x^2$ between $x = 0$ and $x = 4$
Use the table to help you.

x	0	1	2	3	4
y	0				16



[3]

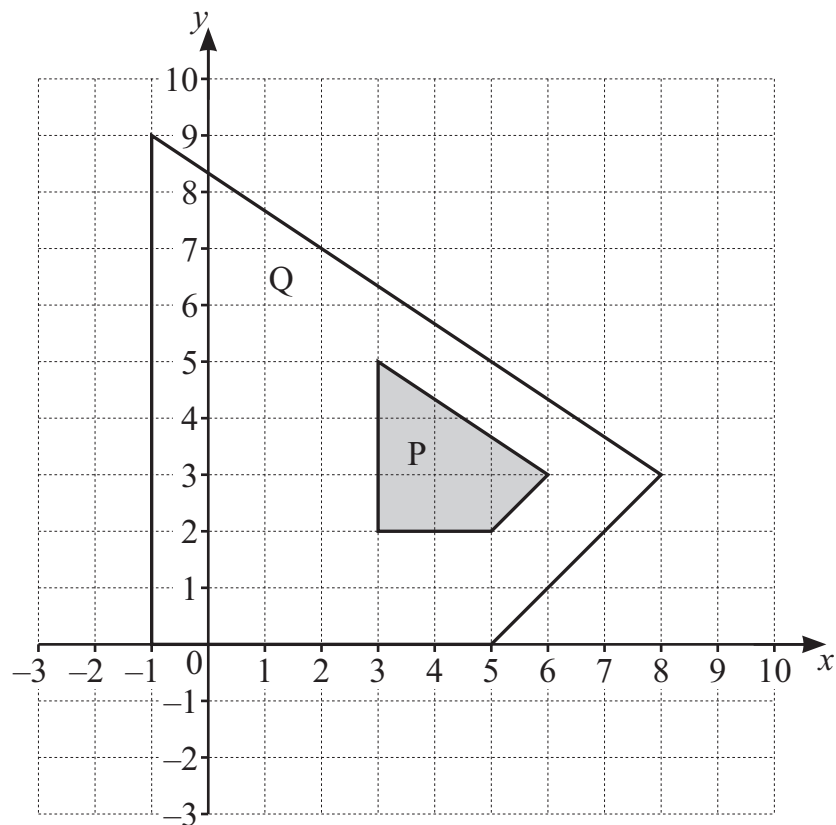
- 14 Sequence A is the linear sequence that begins 3, 6, 9, 12, ...
The n th term for sequence B is $4n + 2$

Tick (✓) to show if each statement is true or false.

	True	False
All terms in sequence B are even numbers.	☐	☐
22 is a term in both sequences.	☐	☐
The numbers that are common to both sequences are multiples of 6	☐	☐

[1]

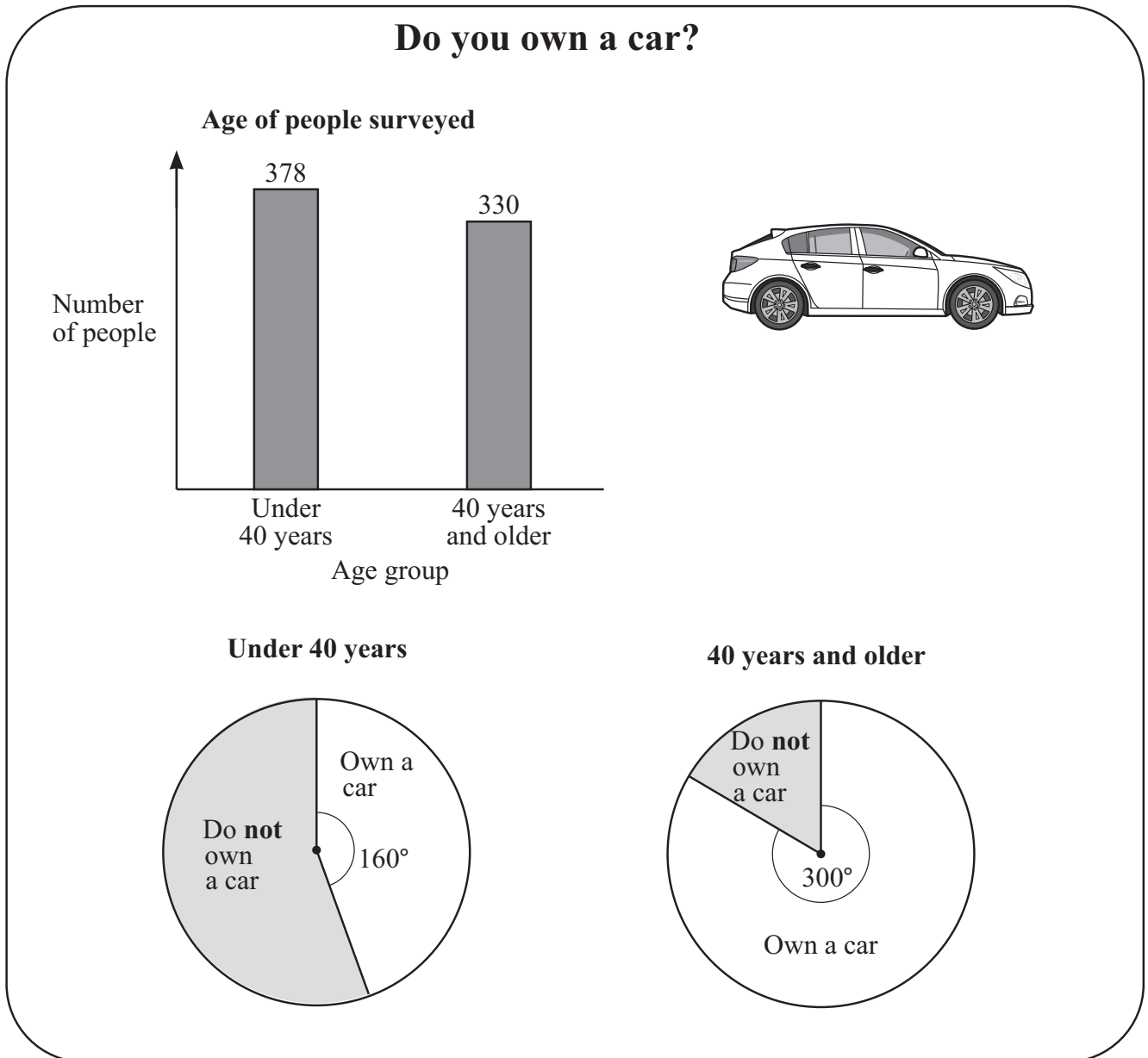
- 15 Two quadrilaterals, P and Q, are shown on the grid.



Describe fully the **single** transformation that maps quadrilateral P to quadrilateral Q.

.....
 [3]

- 16 Oliver draws this diagram to show some information about people he has surveyed.
The diagram shows
the number of people in each age interval,
the proportion of people in each age interval that own a car.



Calculate the total number of people surveyed who own a car.

..... [2]

17 Hassan and Lily share some pens in the ratio 1 : 3

Tick (✓) to show if each statement must be true, could be true or must be false.

	Must be true	Could be true	Must be false
Lily gets more pens than Hassan.	☐	☐	☐
Hassan gets $\frac{1}{3}$ of the pens.	☐	☐	☐
The total number of pens is 20	☐	☐	☐

[2]

18 The circumference of a circle is 36.5 cm.

Find the area of the circle.

..... cm² [3]

19 Here is a function.

$$\begin{array}{ccc} \text{Input} & & \text{Output} \\ x & \longrightarrow & y = 2x^3 \end{array}$$

Complete the table.

Input, x	Output, y
2.5	
	54

[2]

20 Point P has coordinates $(-4, 3)$.
Point P is translated to the point $(x, 0)$, where $x > 0$

Write down a possible vector for this translation.

$\begin{pmatrix} \dots\dots\dots \\ \dots\dots\dots \end{pmatrix}$
[1]

21 Rearrange the formula $p = \frac{3h^2}{5}$ to make h the subject.

$h = \dots\dots\dots$ [2]

- 22** A rectangle has a length of $(8x + 10)$ cm and a width of $(3x + 10)$ cm.
The length of the rectangle is twice the width.

By first writing and solving an equation, show that the area of the rectangle is 1250 cm^2 .

[3]

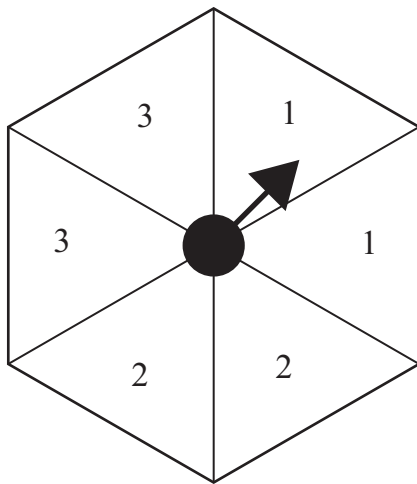
- 23** Gabriella has two fair spinners.
One spinner is blue and the other is red.
She spins both spinners and adds the two results to give a total score.

She says,

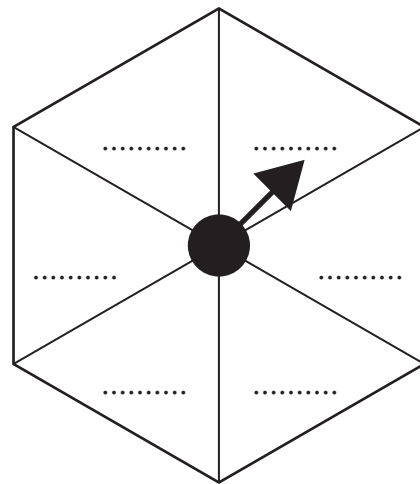
‘Each section of my two spinners is numbered 1 or 2 or 3

The probability that my total score equals 3 is $\frac{1}{9}$ ’

The diagram shows the numbers on the blue spinner.



Blue spinner



Red spinner

Write six possible numbers on the red spinner to make both of Gabriella's statements true.
You may use the table to help you.

Red spinner

Blue
spinner

+						
1						
1						
2						
2						
3						
3						

[2]

24 Here are two ratios.

$$a:b = 2:1 \text{ and } b:c = 4:1$$

Draw a ring around the ratio that is equivalent to $a + b:c$.

12:1

8:1

6:1

3:1

[1]

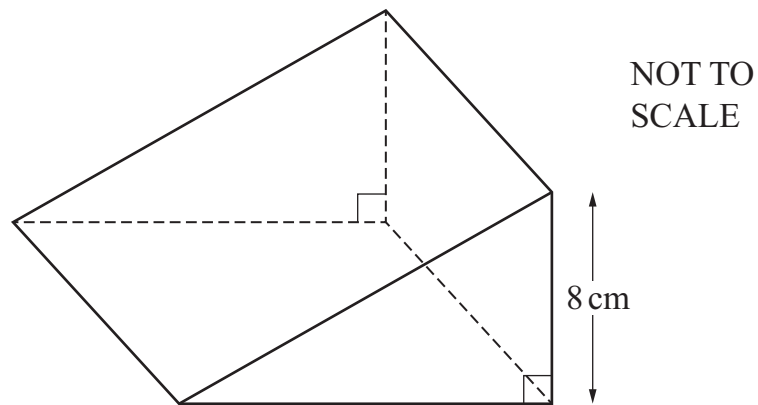
25 Eva can pick 7200 apples in 6 hours.

Yuri picks the same number of apples in 8 hours as Eva picks in **7.5 hours**.

Calculate how many apples Yuri can pick in 9 hours.

..... [2]

26 The diagram shows a solid triangular prism.

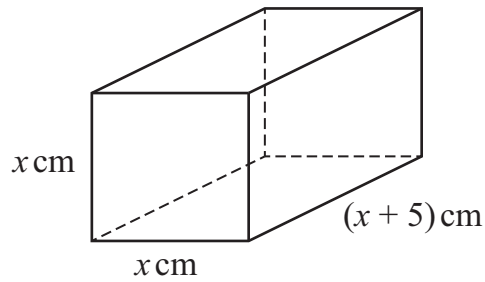
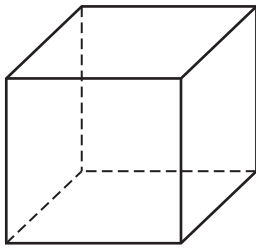


The base of the prism is a square with an area of 225 cm^2 .
The height of the prism is 8 cm.

Calculate the total surface area of the prism.

..... cm^2 [4]

27 The cube and the cuboid have equal volumes.



NOT TO
SCALE

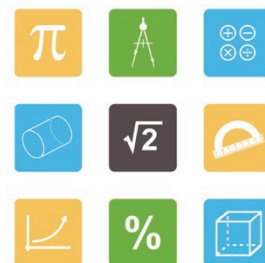
Find an expression for the side length of the cube.

..... cm [2]

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Mathematics

Stage 9

Paper 2

2024

Cambridge Lower Secondary Progression Test

Mark Scheme

PROOF INSTRUCTION (Office use only)	
Checked by (PM Initial)	
Send for /to:	Date
First Proof check	☐ _____
Revision	☐ _____
Vetting	☐ _____
Client	☐ _____
PE / PR Check	☐ _____
Approved for Final Print and Ordering (Initials)	
Printing requirements: Punch hole Yes/No Perforate Yes/No Colour Yes/No Card Yes/No ESM markings Yes/No	

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Square brackets in mark scheme

When square brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6[58...]	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30..... cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e.18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance																																																																						
1	720°	1		Accept any clear indication.																																																																						
2	✓ ✓ ✓	1		All three answers correct for the mark. Accept any clear indication.																																																																						
3	<table><tr><th colspan="4">Choir A</th><th></th><th colspan="5">Choir B</th></tr><tr><td></td><td></td><td>9</td><td>8</td><td>2</td><td>3</td><td>5</td><td></td><td></td><td></td></tr><tr><td></td><td></td><td>5</td><td>3</td><td>3</td><td>1</td><td>3</td><td>4</td><td>4</td><td>9</td></tr><tr><td>7</td><td>4</td><td>1</td><td>0</td><td>4</td><td>2</td><td>5</td><td>6</td><td>7</td><td>7</td></tr><tr><td></td><td>9</td><td>6</td><td>2</td><td>5</td><td>3</td><td>4</td><td>8</td><td></td><td></td></tr><tr><td></td><td></td><td></td><td>3</td><td>6</td><td>0</td><td>7</td><td>8</td><td></td><td></td></tr><tr><td></td><td></td><td>5</td><td>1</td><td>7</td><td>2</td><td></td><td></td><td></td><td></td></tr></table>	Choir A					Choir B							9	8	2	3	5						5	3	3	1	3	4	4	9	7	4	1	0	4	2	5	6	7	7		9	6	2	5	3	4	8						3	6	0	7	8					5	1	7	2					2	Award 1 mark for two or three numbers entered in the correct row (not necessarily ordered).	For 2 marks, the diagram should be ordered.
Choir A					Choir B																																																																					
		9	8	2	3	5																																																																				
		5	3	3	1	3	4	4	9																																																																	
7	4	1	0	4	2	5	6	7	7																																																																	
	9	6	2	5	3	4	8																																																																			
			3	6	0	7	8																																																																			
		5	1	7	2																																																																					
4	Any number x in the interval $344 < x < 345$	1																																																																								
5	$(y =) 6$	1																																																																								
6	$(1, 6)$	1																																																																								
7	60(km/h)	1																																																																								
8	> = < >	2	Award 1 mark for two or three correct symbols.																																																																							

Question	Answer	Marks	Part Marks	Guidance
9	$630(\text{cm}^3)$	2	Award 1 mark for a correct method for finding the area of cross-section or for <i>their</i> area of cross-section $\times 7$	For 1 mark, e.g. $8 \times 10 + 0.5 \times 5 \times 4$ 1 mark can be implied by 90
10	$(\$)5306.04$	2	Award 1 mark for $[5000 \times] \left(1 + \frac{2}{100}\right)^3$ or equivalent	Accept $(\$)5306$ for 2 marks if no wrong working seen. Working may be done in stages, e.g. 5000×1.02 or 5100 and <i>their</i> 5100×1.02 or 5202 and <i>their</i> 5202×1.02
11	8 14	2	Award 1 mark for each correct number.	
12	☒ ☒ ☐	1		Both answers correct for the mark. Accept any clear indication.

Question	Answer	Marks	Part Marks	Guidance
13	A smooth curve which passes through (0, 0), (1, 1), (2, 4), (3, 9) and (4, 16).	3	Award 2 marks for points plotted at four of these points: (0, 0), (1, 1), (2, 4), (3, 9) and (4, 16) with an attempt to join or for all five points plotted correctly with no/incorrect attempt to join. or Award 1 mark for a correctly completed table or for plotting four of <i>their</i> five points correctly.	Implied by correct plots or the remaining three correct coordinates seen.
14	☒ ☐ ☐ ☒ ☒ ☐	1		All three answers correct for the mark. Accept any clear indication.
15	Enlargement and scale factor 3 and centre (5, 3)	3	Award 2 marks for any two out of enlargement, scale factor 3, centre (5, 3). or Award 1 mark for any one out of enlargement, scale factor 3, centre (5, 3).	More than one transformation scores 0 for the question.

Question	Answer	Marks	Part Marks	Guidance									
16	443	2	Award 1 mark for $\frac{160}{360} \times 378$ or equivalent or 168 or $\frac{300}{360} \times 330$ or equivalent or 275										
17	<table><tr><td>☒</td><td>☐</td><td>☐</td></tr><tr><td>☐</td><td>☐</td><td>☒</td></tr><tr><td>☐</td><td>☒</td><td>☐</td></tr></table>	☒	☐	☐	☐	☐	☒	☐	☒	☐	2	Award 1 mark for two correct answers.	Accept any clear indication.
☒	☐	☐											
☐	☐	☒											
☐	☒	☐											
18	An answer in the range 105.6 to 106.2 (cm ²)	3	Award 2 marks for $\pi \times \left(\frac{36.5}{2 \times \pi}\right)^2$ or Award 1 mark for $(r =) \frac{36.5}{2 \times \pi}$ or equivalent or for finding $\pi \times (their\ r)^2$	Accept values of π between 3.14 and $\frac{22}{7}$ Implied by $(r =)$ 5.8 to 5.8122									
19	<table><tr><td>2.5</td><td>31.25</td></tr><tr><td>3</td><td>54</td></tr></table>	2.5	31.25	3	54	2	Award 1 mark for each correct value.	Accept equivalent fractions for 31.25, e.g. $\frac{125}{4}$					
2.5	31.25												
3	54												

Question	Answer	Marks	Part Marks	Guidance
20	A vector of the form $\begin{pmatrix} a \\ -3 \end{pmatrix}$ where $a > 4$	1		e.g. $\begin{pmatrix} 5 \\ -3 \end{pmatrix}$ or $\begin{pmatrix} 4.1 \\ -3 \end{pmatrix}$
21	$(h =) [\pm] \sqrt{\frac{5p}{3}}$ or equivalent	2	Award 1 mark for a correct first step, e.g. $5p = 3h^2$ or better $\frac{p}{3} = \frac{h^2}{5}$ or better $\sqrt{p} = \sqrt{\frac{3h^2}{5}}$ or better.	Or equivalent for 2 marks, e.g. $\sqrt{5p} \div \sqrt{3}$
22	Correct demonstration that area is $1250(\text{cm}^2)$. Answer should show evidence of - a correct equation $(x =) 5$ - a correct method for finding the area, e.g. $(8 \times 5 + 10) \times (3 \times 5 + 10)$ or 50×25	3	Award 1 mark for a correct equation, e.g. $8x + 10 = 2(3x + 10)$ or better $\frac{8x + 10}{2} = 3x + 10$ or better and Award 1 mark for $(x =) 5$ or $(8 \times \text{their } 5 + 10) \times (3 \times \text{their } 5 + 10)$ or better.	Correct equation is not implied by $x = 5$ alone.

Question	Answer	Marks	Part Marks	Guidance
23	Spinner numbered 1, 2, 3, 3, 3, 3 or 2, 2, 3, 3, 3, 3 or 1, 1, 3, 3, 3, 3	2	Award 1 mark for $\frac{1}{9} = \frac{4}{36}$ or for writing exactly four 3's in the interior of the sample space diagram or for numbering spinner to make $P(\text{total} = 3) = \frac{1}{9}$ but including numbers other than 1, 2 or 3	For 2 marks, accept answer written in the column headings of the sample space diagram if spinner is not completed. e.g. 1, 2, 3, 3, 4, 4 is 1 mark.
24	12 : 1	1		Accept any clear indication.
25	10 125	2	Award 1 mark for $[7200 \times] \frac{7.5}{6} \times \frac{9}{8}$ or equivalent or [number of apples Yuri picks in 8 hours =] $\frac{7200}{6} \times 7.5$ or 9000 or $\frac{\text{their } 9000}{8} \times 9$	

Question	Answer	Marks	Part Marks	Guidance
26	$720(\text{cm}^2)$	4	Award 1 mark for (square length =) 15 (cm) and Award 1 mark for $8^2 + (\text{their } 15)^2$ or better and Award 1 mark for $2 \times 0.5 \times \text{their } 15 \times 8$ or better or $\text{their } 15 \times 8 + \text{their } 17 \times \text{their } 15 [+ 225]$ or better.	Or better, e.g. $\sqrt{8^2 + (\text{their } 15)^2}$ or 17 Or better, e.g. 120 Or better, e.g. 375 or 495 or 600
27	$\sqrt[3]{x^2(x+5)}$ Final answer or equivalent	2	Award 1 mark for $x^2(x+5)$ or equivalent or $\sqrt[3]{\text{their } x^2(x+5)}$	Or equivalent, e.g. $\sqrt[3]{x^3 + 5x^2}$ Or equivalent, e.g. $x^3 + 5x^2$

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